METRIC

DOD-STD-1685(SH)
10 October 1978

MILITARY STANDARD

COMPREHENSIBILITY STANDARDS FOR TECHNICAL MANUALS

(METRIC)

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DOD-STD-1685(SH)
10 October 1978

DEPARTMENT OF DEFENSE

Washington, DC 20362

Comprehensibility Standards for Technical Manuals.

DOD-STD-1685(SH)

1. This standard is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

1. Maximum comprehensibility of technical data is vital in maintaining operational readiness of systems and equipment, in reducing life cycle costs, and in increasing the probability of successful repair on board ship. Technical manuals have often been produced with style, format, and organization characteristics which impose an unnecessary burden on the user in terms of reading, studying, and conceptualizing skills.

2. The requirements in this standard are intended to reduce this burden upon the user and to foster a user-oriented approach by the technical manual preparer. This approach consists of a greater sensitivity to the characteristics and needs of the user. It should result in a greater awareness of the possibilities for presenting technical data in ways that directly support the performance of technical tasks.

3. This standard reflects only a subset of the characteristics of user-oriented technical manuals. Many other characteristics of user-oriented manuals cannot now be specified or measured. Conformance with the requirements in this standard will improve comprehensibility and will foster, but not guarantee, a user-oriented approach.

4. In cases where sampling is to be used in quality assurance procedures, a simple random sampling of material from a technical manual is unsatisfactory for two reasons: (a) the material in the manual is not a homogeneous mass; and (b) the number of different characteristics to be checked is large. An improved sampling plan is presented in an appendix.

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1. GENERAL, PURPOSE, OBJECTIVE, AND APPLICATION

1.1 General. This standard sets forth style and format requirements related to technical manual comprehensibility.

1.2 Purpose. The purpose of these requirements is to improve the match between how technical information is presented and how technical manual users comprehend the information.

1.3 Objective. Technical manuals produced in accordance with these requirements will have the following characteristics: familiar words, uniform vocabulary, short sentences, short paragraphs with informative headings, organization of material based on the immediate needs of the user, and a large number of simple, clear illustrations.

1.4 Application. This standard is applicable to procurements of new technical manuals, update or complete revisions, supplements, and technical manual changes.

2. REFERENCED DOCUMENTS

2.1 Not applicable.

3. DEFINITIONS

3.1 Average sentence length (ASL). A measure of readability computed by dividing the number of words by the number of sentences in the text. The ASL for the whole text is often estimated by using samples of text.

3.2 Average word length (AWL). A measure of readability expressed either as syllables per word or keystrokes per word. To get syllables per word, the number of syllables (as the word is pronounced) is divided by the number of words in the text. Keystrokes per word is used when there is a device to count keystrokes automatically as the text is typed. The AWL for the whole text is often estimated by using samples of text.

3.3 Callout. In pictorials of equipment assemblies or subassemblies, parts or features may be identified by a symbol called a callout. The callout is located outside the boundaries of the pictured equipment with a line (leader) or arrow connecting the symbol to the part or feature. The symbol may be the name of the part or feature (nomenclature callout), or it may be a number keyed to a legend or to explanatory text (number callout).

3.4 Clutter. In an illustration, a condition in which the amount of information in a given area is so great that confusion and errors of interpretation occur.

3.5 Comprehensibility. How well information can be located, understood, remembered, or acted upon by the user. High comprehensibility permits the user to have a minimum of searching, reading, studying, and conceptualizing skills.

3.6 Consolidation. Organization of material so that all information in the manual necessary to complete a single task is in one place.

3.7 Function diagram. A type of illustration (see 3.8) in which symbols are connected by lines to show relationships among the symbols. The symbols may be rectangles or other shapes, standard symbols representing components or functions, or pictorials representing equipment or components. The lines may represent physical things, such as wires, or ideas, such as signal or logic flow. This category includes schematics, wiring and piping diagrams, logic diagrams, flow charts, and block diagrams (see figure 1).

3.8 Illustration. A general term meaning graphic presentations of all types. Illustrations include pictorials, function diagrams, and line graphs. This term is used instead of such terms as figure, graphic, drawing, diagram, and artwork. It refers to all material in technical manuals titled "Figure . . . ." (see figure 1).

3.9 Line graph. A type of illustration (see 3.7) which shows one or more numeric relationships between two variables. Values of the variables are indicated on the horizontal and vertical axes. The relationships between the axis variables are indicated by lines (curves, straight line segments, bars, or other devices as applicable) (see figure 1).

3.10 Nomenclature (equipment). The official name or alphanumeric identifier for an equipment or component.

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3.11 Paragraph. A division of nonprocedural text (see 3.18).

3.12 Pictorial. A type of illustration (see 3.8) showing the physical appearance of an equipment or component parts. This term is used instead of such general terms as illustration, drawing, and diagram (see figure 1).

3.13 Procedure. A list of steps giving the sequence of actions which make up a task (see 3.18).

3.14 Readability. A type of text comprehensibility measured by such variables as word length, sentence length, paragraph length, and procedural step length.

3.15 Relevance. Given a limited segment of technical material, such as a single procedure, paragraph, or illustration, all information in the segment relates directly to the purpose or theme of the segment.

3.16 Task (user). A sequence of user actions with a beginning and an end. User tasks relate to installation, checkout, maintenance, and operation of systems or equipment. In addition, in this standard, training, learning, review, and search for information are also considered types of user tasks.

3.17 Technical contents sections of technical manual. Sections of a technical manual whose purpose is to guide actions of the user or to provide background for guiding actions of the user in performing technical tasks. This excludes front matter, indexes, and rear matter.

3.18 Text. Text refers to the written parts of the technical sections excluding labels, legends, and callouts in illustrations. This standard defines two kinds of text: procedural and nonprocedural. Procedural text specifically directs the actions of the user. Other text, such as explanation, theory, and description, is called nonprocedural text.

3.19 User. A person supported by the technical manual in performing technical tasks.

4. GENERAL REQUIREMENTS

4.1 Support of user tasks. In all cases to the extent possible, information shall be presented in a way which:

(a) Directly supports the immediate task of the user.
(b) Minimizes demands on his reading, studying, and conceptualizing skills.
(c) Minimizes his need to refer to other documents or text material.

4.2 Organization of the technical manual. The technical manual shall be organized on the basis of the user's need for information as he performs his tasks. In complex equipment or systems, where hierarchies of functional and physical relationships exist, corresponding technical tasks shall similarly be organized in hierarchies. An overview of the hierarchical arrangement of tasks by which the manual is organized shall be provided. This may be illustrated as a family tree or top-down breakdown of the tasks. The hierarchical arrangement shall be described in the introductory chapter of the technical manual.

4.3 Consolidation of material. To the extent possible, all information required to complete a single task shall be put in one place. Information and artwork shall be repeated as necessary to fulfill this requirement. The need for the user to refer to other parts of the manual or to other information sources during performance of a task shall be minimized.

4.4 Text.

4.4.1 Vocabulary. The simplest, most familiar, and most concrete words which accurately convey the intended meaning shall be used. Short words, words typically learned early in life, and words used in ordinary conversation shall be preferred. Highly technical terms shall be limited to circumstances where simpler terms would not accurately convey the intended meaning. Highly technical terms shall be defined directly after their use or in a glossary.

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4.4.1.1 Concrete and specific language. Concrete and specific language shall be used to reduce vagueness. Examples:

(a) Abstract and general: Precautions should be taken against fire.
Concrete and specific: Periodically inspect fire doors and automatic sprinkler system.

(b) Abstract and general: It is wise to make the area comfortable and to make provisions for shutting out noise from the shop.
Concrete and specific: Air-condition the area and soundproof the walls.

4.4.1.2 Simple versus elaborate phraseology. Unnecessarily elaborate phraseology shall be avoided. Examples:

Simple	Unnecessarily elaborate
About	As to, with regard to
After	Subsequent to
Because	Inasmuch as, for the reason that
Require	Necessitate
Enough	Sufficient
Find out	Ascertain
For	For the purpose of
Get	Acquire, obtain, receive
If	In the event that
Now	At this time, presently
Since	Inasmuch as

4.4.1.3 Variety. The use of synonyms to provide variety in technical writing can lead to confusion. If there are synonyms for a concept or object, one of them shall be selected and used consistently throughout the technical manual. For example, do not use the terms unit, assembly, equipment, component, and hardware interchangeably to refer to the same item.

4.4.1.4 Preferred verb usage. Appendix A contains a list of verbs commonly used in technical manuals. For many of these, preferred alternatives are given. The verbs in the PREFERRED ALTERNATIVES column shall be used in preference to corresponding verbs in the VERBS column unless the meaning is distorted or the sentence becomes awkward. Nonspecific verbs shall be avoided in favor of verbs designating specific user actions.

4.4.1.5 Nomenclature. Nomenclature associated with systems, equipment, and components shall be introduced by means of pictorials coordinated with descriptive text.

4.4.1.6 Shortened equipment names. In each section supporting a single user task, official terminology shall be used for the first reference to a hardware item. Official terminology shall be used wherever the use of a shortened name might be ambiguous.

4.4.1.7 Abbreviations and acronyms. Abbreviations and acronyms shall be used sparingly. They shall be defined on the first use in each section relating to a single user task. The use of abbreviations and acronyms shall be such that there is no need for experienced users to consult a glossary of abbreviations and acronyms. Abbreviations and acronyms should be used for those systems, subsystems, equipments, components, organizations or other items which are ordinarily referred to by their abbreviations or acronyms by experienced personnel. When abbreviations or acronyms are used as markings on the equipment, the same abbreviations and acronyms shall be used in the technical manual.

4.4.2 Sentence structure and grammar.

4.4.2.1 Word order. Sentences and clauses shall be written using simple word order (subject, verb, object) to the extent possible. Modifiers, including prepositional phrases, shall be as close as possible to the word modified. These word order requirements shall be relaxed only to the extent necessary to avoid ambiguity or distortion of meaning.

4.4.2.2 Complex sentences. If a sentence has more than one clause and is more than 20 words in length, it shall be rewritten as one or more simple sentences if possible.

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4.4.2.3 List form. If any series of three items or more appears in a sentence, the sentence shall be written so the items will appear in explicit list form. Example:

All dirty, greasy, or water-repellent surfaces are then rubbed or scrubbed with swabs or brushes.

Using swabs or brushes, rub or scrub all surfaces which are:
dirty
greasy
water-repellent.

4.4.2.4 First person pronouns. First person pronouns may be used, if applicable, to avoid indefinite passive construction, for example, "We found that..." rather than "It has been found ....".

4.4.2.5 Standard English grammar. Rules of grammar for standard American English shall be used. Colloquial, substandard, and slang expressions shall not be used. Infinitives may be split and sentences may be ended with prepositions in cases where not doing this sounds awkward or stilted, and is likely to distract the user's attention from comprehending the content.

4.4.2.6 Procedural text.

4.4.2.6.1 Sentences in procedures. Sentences directing the actions of the user shall be in the active voice and imperative mode. If more than one person is involved, the directions shall be in the active voice, indicative mode, third person (for instance, "Man A shall ... while Man B ...").

4.4.2.6.2 Positive form. Procedural steps shall be in positive form ("Close the container" rather than "Do not leave the container open") unless the meaning demands the negative form.

4.5 Illustrations and tables.

4.5.1 Dependence on illustrations. Illustrations shall be considered at least as important as text in conveying information. To the extent possible, text shall reference supporting illustrations to carry the main burden of conveying information. The text shall appear either as part of the illustration or separately. Number or nomenclature callouts shall be used to key the important features of the illustration to the text.

4.5.2 Nearness of illustrations and text. Where possible, text and related illustrations shall be arranged so they can be viewed at the same time. If this is not possible (for example, more than one full-page illustration), the illustrations shall be placed as soon after the first reference as possible. If the same illustration is referred to in procedural steps on a number of pages, the illustration shall be repeated as often as necessary.

4.5.3 Relevance of illustrations. Each illustration shall contain only that information directly relevant to a specific user task. The same illustration shall not be used for more than one task unless everything in the illustration relates to each task. The intent is to provide several small, simple illustrations rather than a single, large, complex one containing information not relevant to the immediate task.

4.5.4 Illustrations per page. The fraction obtained by dividing the total number of illustrations and tables by the number of pages in the technical contents sections of the manual shall be 0.50 or greater.

4.5.5 Titles of illustrations and tables. Each illustration or table shall have a title. The title shall identify the contents or purpose of its illustration or table and distinguish that illustration or table from others in the technical manual. Each title shall identify the variables represented in its illustration or table.

5. DETAILED REQUIREMENTS

5.1 Procedure and paragraph headings. (Note that the terms "procedure" and "paragraph" are not identical--see 3.11 and 3.13.) Headings shall be unique within major sections of the technical manual.

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5.1.1 Procedure headings. Every procedure shall have a heading.

5.1.2 Paragraph headings. The ratio of paragraphs to paragraph headings shall not exceed 2:1.

5.1.3 Informative headings. Headings shall be informative and not merely labels. Headings shall serve two purposes. First, they shall make access to information easier by permitting the reader to scan a page to find the information he wants. Second, they shall alert the reader to the type of information contained in the text to follow. Examples:

Label heading	Informative heading
Procedure	Procedure for recording temperature
Instrumentation	Heat measured by thermocouples
Discussion	Reasons for waveform variations
General description	Symbols created for radar PPI

5.1.4 Headings and relevance. A paragraph or procedure heading shall uniquely identify the content of the material which it heads. All material under a single heading shall be consistent with the heading.

5.1.5 Heading length. Not more than 10 percent of all headings shall be longer than 7 words. No heading shall be longer than 10 words.

5.1.6 Headings per page. To facilitate the user's search for information, in the technical contents sections of the manual:

(a) If more than half of a page consists of text, there shall be at least two headings on the page.
(b) If one-fourth to one-half of a page consists of text, there shall be at least one heading on the page.

5.2 Typography.

5.2.1 Typography in text. Typography in text shall be as follows:

(a) Minimum type size on the final printed material shall be 10-point (3.5 millimeters (mm), 0.14 inches), except for parts lists, which may be 8-point, (2.8 mm, 0.11 inches).
(b) Minimum leading size shall be 1-point (0.35 mm, 0.014 inches) if column width is 40 characters or less, or 2-point (0.70 mm, 0.28 inches) if column width is over 40 characters.
(c) Upper and lower case shall be used except for major section titles and acronyms, which shall be upper case.
(d) Common, ordinary typefaces, not stylish or condensed fonts, shall be used.
(e) Paragraph and procedure headings shall be readily distinguishable from body of text.
(f) Black ink on non-glossy white paper shall be used to the extent possible.

5.2.2 Typography in illustrations. Minimum type size on the final printed material shall be 8-point (2.8 mm, 0.11 inches), boldface type, all capitals. When 10-point (3.5 mm, 0.14 inches) or larger type will be the final size, the boldface requirement may be dropped.

5.3 Readability. Measurements cited in these readability requirements shall be based either on the technical contents sections of the entire technical manual, or on samples obtained by the method of Appendix C. AWL and ASL shall be calculated using one of the procedures described in Appendix B.

5.3.1 AWL. AWL shall not exceed 1.6 syllables per word, or 5.1 strokes per word, depending on the method of calculation used.

5.3.2 ASL. ASL shall not exceed 15 words.

5.3.3 Maximum sentence and paragraph lengths in nonprocedural text. No more than 5 percent of all sentences shall exceed 20 words. No more than 5 percent of all paragraphs shall exceed 60 words.

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5.3.4 Step length in procedures.

5.3.4.1 Words. No more than 5 percent of all procedural steps shall exceed 25 words; no procedural step shall exceed 30 words.

5.3.4.2 Sentences. No more than 5 percent of all procedural steps shall exceed three sentences. No procedural step shall exceed four sentences. No sentences in procedural steps shall exceed 15 words.

5.4 Procedures.

5.4.1 Form of procedures.

5.4.1.1 Procedural step content.. A procedural step shall be limited to a single operation or to repetitions of a single operation with the following exceptions:

(a) If simultaneous operations are required, they shall be listed together in the same step.
(b) If the step is a major step in a dual-level presentation (see 5.4.1.3), the major step shall express the action with a single verb, for instance, "Turn equipment ON."
(c) If the step represents a detailed procedure, as in (b), but the procedure is so basic that the details should rarely be needed by the intended users, or if the procedure is very frequently repeated, such as turn on, turn off, and calibrate procedures, the step shall refer to the procedure with a single verb and cite a reference to the detailed steps; for instance, "Turn equipment ON (see 3-46 for details)." (See 5.4.5 and 5.4.2).
(d) Verification of the result of an operation may be included in the step, for example, "Press pushbutton A and be sure indicator A is lit."

5.4.1.2 Procedure length. Each separate procedure shall contain no more than ten steps.

5.4.1.3 Dual-level presentation. If users of a procedure are expected to vary widely in experience and capability, the procuring agency may direct that certain procedures be presented in a dual-level format. In this format, detailed steps appropriate for inexperienced users shall be listed under major steps. The major steps, without reference to the detailed steps, shall be sufficient to support the performance of experienced users (see figure 2).

5.4.2 Introduction to procedure. At the beginning of a procedure, prior to the listing of procedural steps, introductory information shall be provided to enable the user to carry out the procedure without interruption. Examples of relevant introductory information:

(a) References to relevant explanatory, descriptive, or theoretical material.
(b) Number and qualifications of personnel required to carry out the procedure.
(c) Test equipment required.
(d) Special tools required.
(e) Supplies and materials required.
(f) Condition of the equipment at the beginning of the procedure, and references to procedures required to establish this condition.
(g) List of any steps in the procedure which are themselves procedures detailed elsewhere in the manual, with references to these detailed procedures.
(h) Warning, if applicable, to note and obey safety notices in the text.

5.4.3 Lead-in. Procedural steps shall not be prefaced by a lead-in which merely duplicates the heading as in the following example:

"Disassembly of a sensing unit. The sensing unit is disassembled according to the following procedure: . . ."

(See Appendix A; a number of nonspecific words are frequently used in such lead-ins.)

5.4.4 Relevance of procedural text. A procedure shall present only information necessary for completion of the task or directly helpful in preventing error.

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5.4.5 Consolidation and references. All information necessary to carry out a procedure shall be consolidated in one place. Once the sequence of actions is begun, there shall ordinarily be no need to refer to other parts of the manual for information necessary to continue the procedure. In cases where a procedural step is itself a procedure described elsewhere in the manual (for instance, instructions for turning equipment on), reference may be made to the detailed procedure (see 5.4.1.1). This reference shall also be cited in the introduction to the procedure (see 5.4.2), so that the user may locate the referenced material before starting the procedure.

5.4.6 Confirming events. Confirming events are information not essential to performing the task but helpful to the user. They confirm the correctness of his actions and confirm that the equipment is operating properly. Such confirming events shall be included in procedural steps. For example, if as a result of performing a procedural step a motor starts or an indicator light goes on, such information shall be included.

5.4.7 Logic trees and decision tables. Troubleshooting procedures involving branches or multiple alternatives shall be supported by graphic presentations such as logic trees, decision tables, or Maintenance Dependency Charts (MDCs). Decision points in the procedure which are not inherently binary (two-choice) shall not be forced to appear as a series of binary choices in the logic tree or decision table.

5.5 Nonprocedural text.

5.5.1 Topic sentence. The first sentence of each paragraph shall be the topic sentence. The topic sentence shall identify or summarize the content of the paragraph. All information in the paragraph shall relate to it.

5.5.2 Separation of procedural and nonprocedural text. Explanatory, descriptive, or theoretical paragraphs shall not contain procedures, and procedures shall not contain explanatory, descriptive, or theoretical material.

5.6 Illustrations.

5.6.1 Function diagrams. (See 3.7 and 3.8 for definitions.)

5.6.1.1 Clutter in function diagrams. There shall be no more than 120 components on any page (page unit of a foldout). The number of line intersections allowable in a 51 by 51 millimeter (mm) (2 by 2 inch) area shall depend on the number of components in the same area. The number of intersections plus twice the number of components shall be 30 or less (see figure 3).

5.6.1.2 Component identity. Each component shall be identified. Where applicable, the wording on the diagram shall correspond exactly with the wording on the equipment.

5.6.1.3 Inputs and outputs. All inputs and outputs shall be clearly identified. To the extent possible, they shall be placed near the diagram edge, with inputs at the left or top, and outputs at the right or bottom.

5.6.1.4 Direction of signal flow. Where possible, major signal or pressure flow shall be from left to right and feedback or return flow shall be from right to left. The direction of flow shall be indicated by arrows.

5.6.2 Wiring and piping diagrams.

5.6.2.1 Line layout. In single page diagrams, termination points shall be shown for every wire or pipe. In multipage diagrams, unterminated line segments shall be identified by appropriate symbols with references maintaining continuity from page to page. The diagrams shall be laid out to eliminate jogs in lines where possible, and to minimize the number of crossing lines. There shall be no more than 20 line crossings (whether or not they indicate physical connections) in any 51 by 51 mm (2 by 2 inch) area.

5.6.2.2 Line weights. Heaviest line weights shall be used for the outlines of the components. Lighter line weights shall be used to indicate wire or pipe circuits.

5.6.2.3 Circuit identification. Lines representing wires, pipes, and their segments shall be identified with the applicable identifying number and color code.

5.6.2.4 Component identification. Components shown on the diagram shall be identified by nomenclature and reference designation.

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5.6.2.5 Table of components associated with wiring diagrams. A table of components shall list the reference designation, nomenclature, location, and access cover, if applicable, for every component on the diagram. This table may be on a separate page.

5.6.3 Pictorials. (See 3.8 and 3.12 for definitions.)

5.6.3.1 Line drawings versus halftones. Line drawings shall be used to eliminate irrelevant detail, emphasize relevant information, and allow flexibility in viewpoint and scale. Line drawings shall be used instead of halftones such as photographs and airbrush drawings.

5.6.3.2 Plane of view. The view of objects in pictorials shall be identical to the view the user sees when performing his task.

5.6.3.3 Locator views. When the location of the illustrated part is not obvious, a locator view with directional arrows shall be provided. It shall show the orientation of the view with respect to the next higher assembly (see figure 4).

5.6.3.4 Axis lines. The relationship of all parts in an exploded view shall be clearly indicated by axis lines.

5.6.3.5 Callouts.

5.6.3.5.1 Placement of callouts. All callouts shall be placed outside the boundaries of the parts illustrated.

5.6.3.5.2 Callout number sequence. Callout numbers shall be assigned clockwise in sequence. The sequence shall begin at the upper left of the pictorial unless there is a more natural place to begin, such as the end of an object rather than the middle. Callout numbers shall not be duplicated in the same illustration.

5.6.3.5.3 Callout leaders. Callout leaders or arrows shall be straight lines where possible. They shall not cross each other.

5.6.3.5.4 Clutter from callouts. Callouts shall be clearly legible and easily distinguishable from each other. Callouts and their leader lines shall be easily distinguishable from components and other lines of the illustration (see figures 5 and 6). Number callouts should be circled if the user's ability to locate the numbers would be improved.

5.6.4 Line graphs. (See 3.8 and 3.9 for definition.)

5.6.4.1 Clutter in line graphs. The number of ideas conveyed per graph shall be minimized. Line graphs shall depict a maximum of four relationships between the axis variables. Lines depicting relationships shall be coded for easy discrimination.

5.6.4.2 Orientation of axes. If there is a natural orientation for the axes, for example, altitude on the vertical axis, the axes shall be so oriented.

5.6.4.3 Grid lines. The number of grid lines used shall be such that the user can read values to the required degree of accuracy. Size of the illustration shall be such that the grid lines shall be no less than 0.1 inches apart. Grid lines shall be lighter than the graph lines, and shall not obscure detail necessary for proper use of the graph.

5.6.4.4 Graph scales. Graph scales may be linear or nonlinear as required for proper comprehension and use. The axes shall be labelled to indicate the variables and units of measurement.

5.6.4.5 Instructions for use. For complex graphs, instructions shall be provided for use and interpretation.

5.6.5 Tables.

5.6.5.1 Units of measurement in tables. Units of measurement shall be specified in row or column headings, as appropriate.

5.6.5.2 Interpolation. No interpolation (that is, estimation between tabled values) shall be required of the user. Interpolation shall be avoided by expanding the table or by presenting the data in a graph.

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5.6.5.3 Alinement of entries in tables. Entries shall be alined within columns as follows:

(a) For decimal data, decimal points shall be alined.
(b) For scientific notation, multiplication signs shall be alined.
(c) All other numeric data shall be alined flush right.
(d) Alphabetic or alphanumeric data shall be alined flush left.

5.6.5.4 Groupings of rows in tables. The row entries in tables shall be arranged in groups of up to five rows. If all cells in a table have entries, groups of rows shall be separated with clear space. If some entries are missing, light horizontal lines shall be used to separate the groups.

5.6.5.5 Clutter in tables. To avoid clutter, at least 25 percent of the area within the table shall be clear space between columns and groups of rows.

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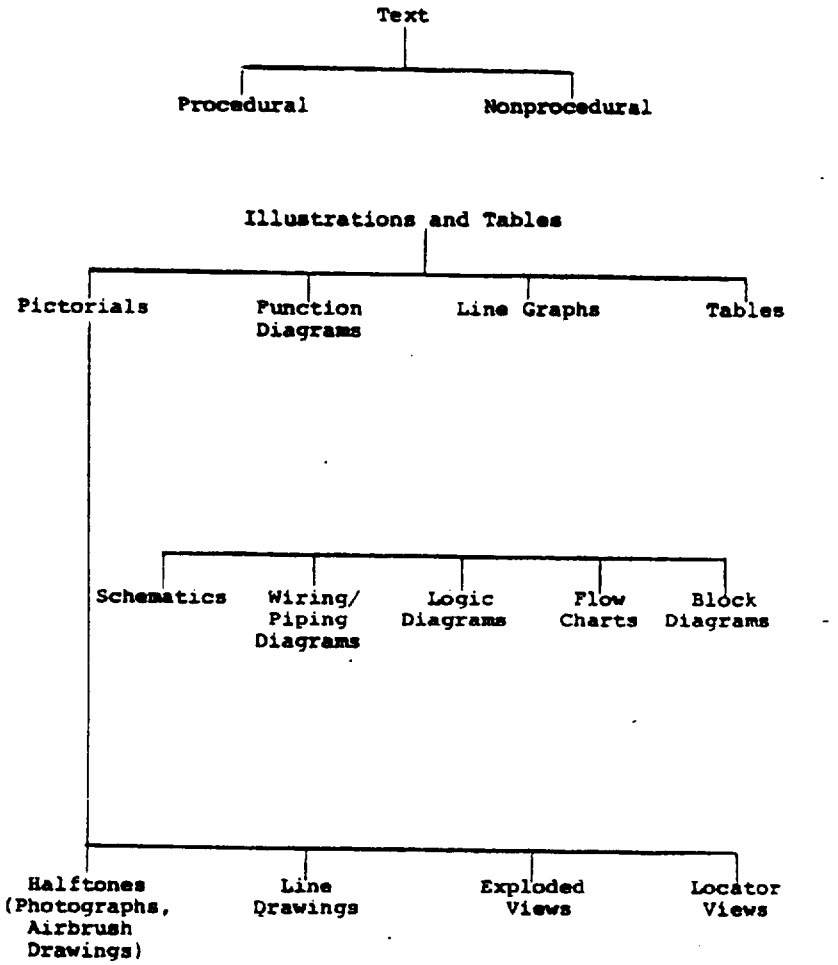


FIGURE 1. Breakdown of presentation types covered by this standard.

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1. Remove air cleaner hose.

   a. Using blade screwdriver, loosen lower clamp screw.
   b. Spread clamp until it is loose.
   c. Slide off air cleaner hose.

2. Remove vacuum line.

   a. Using 5/16 inch wrench, unscrew vacuum line nut.
   b. Pull vacuum line out of carburetor.

3. Remove throttle linkage.

   a. Using 7/16 inch wrench, unscrew throttle link nut.
   b. Pull throttle link from carburetor.

4. Remove fuel line.

> CAUTION
>
> When fuel line is pulled away, fuel may leak onto engine. Be sure engine is cool before removing fuel line.

   a. Using 5/8 inch wrench, unscrew fuel line nut.
   b. Pull fuel line from carburetor.

5. Remove choke linkage.

   a. Using 3/8 inch wrench, loosen cable housing clamp nut.
   b. Using 3/8 inch wrench, loosen cable clamp nut.
   c. Slide cable housing and cable out and away from carburetor.

6. Remove carburetor.

   a. Using 5/8 inch wrench, unscrew mounting nuts to the top of the mounting studs.
   b. Lift carburetor, then unscrew and take off mounting nuts.
   c. Lift carburetor off of mounting studs.

FIGURE 2. Example of dual-level presentation procedure.

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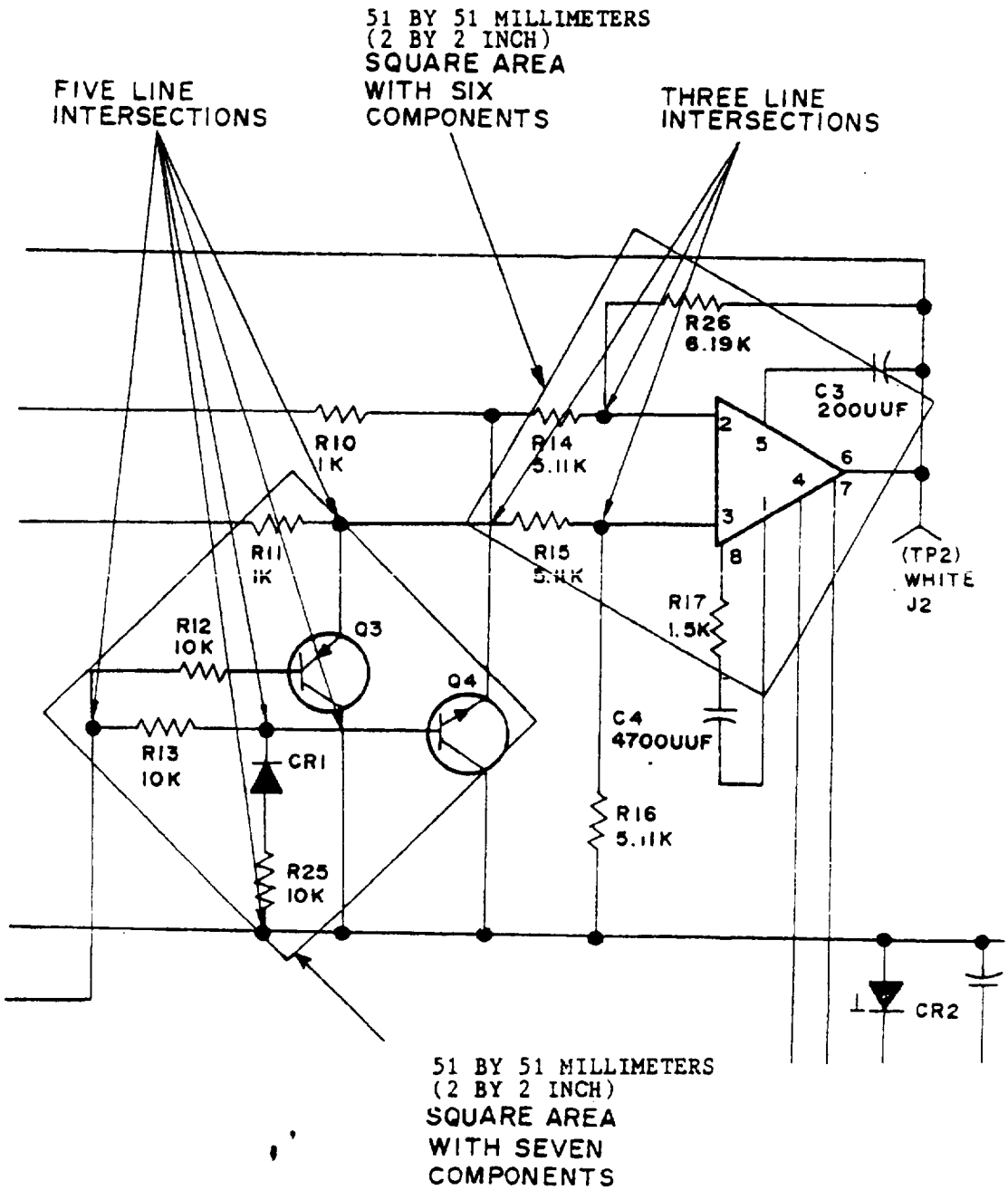


FIGURE 3. Evaluation of clutter in a function diagram (schematic).

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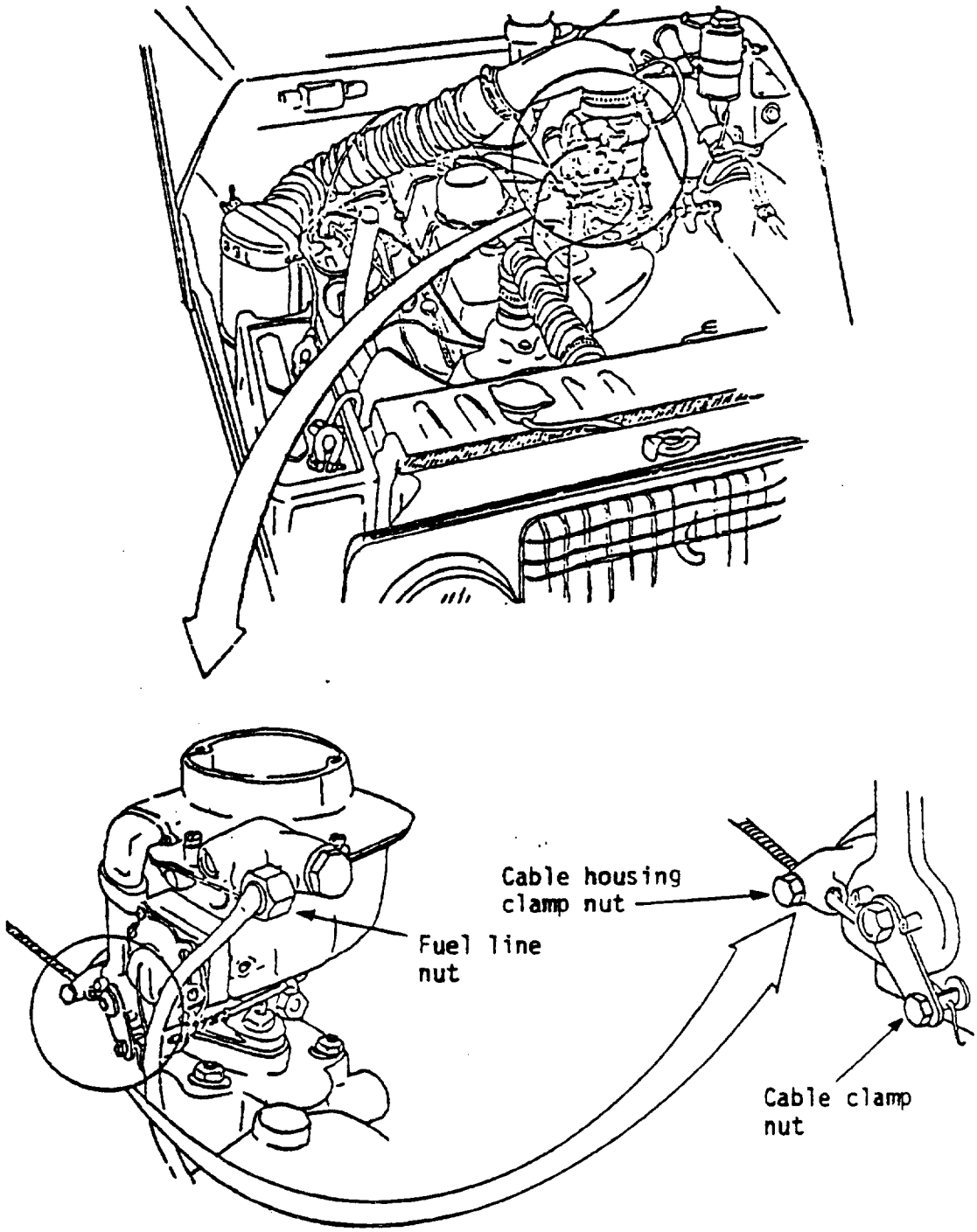
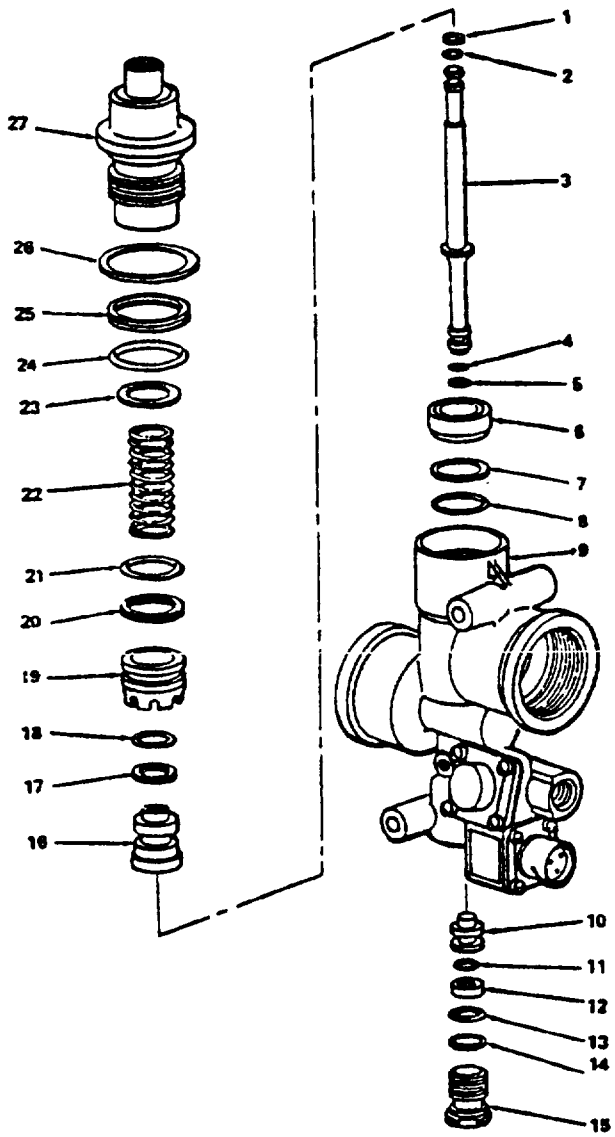


FIGURE 4. Example of use of pictorials to introduce nomenclature and example of locator view.

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1. BACKUP RING
2. O-RING
3. ACTUATOR
4. O-RING
5. BACKUP RING
6. SEAT
7. BACKUP RING
8. O-RING
9. BODY
10. PISTON
11. O-RING
12. TEFLON RING
13. O-RING
14. BACKUP RING
15. PLUG
16. POPPET
17. BACKUP RING
18. O-RING
19. GUIDE
20. BACKUP RING
21. O-RING
22. SPRING
23. SHIM
24. O-RING
25. BACKUP RING
26. SHIM
27. END CAP ASSEMBLY

FIGURE 5. Example of good callout placement.

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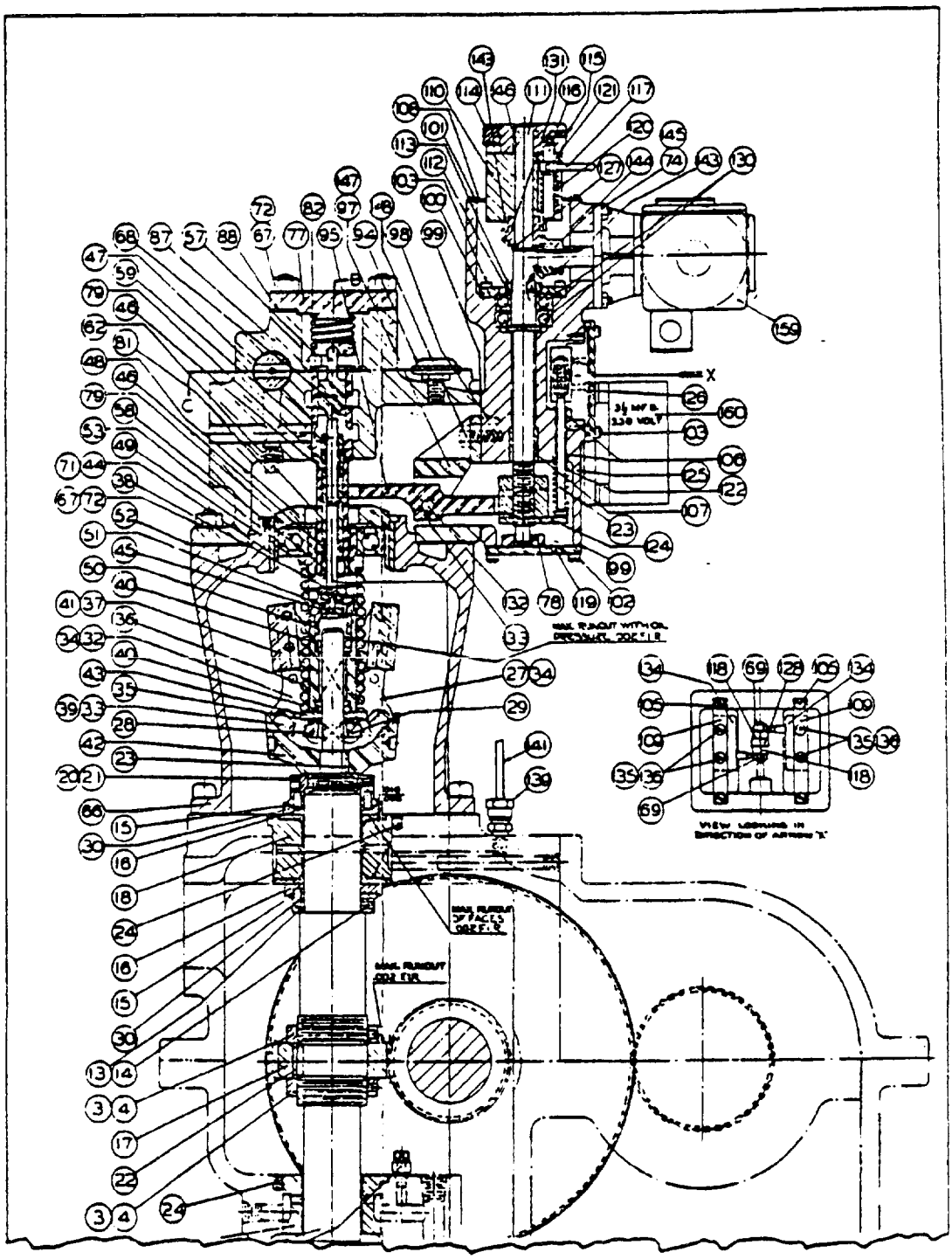


FIGURE 6. Example of clutter due to number callouts.

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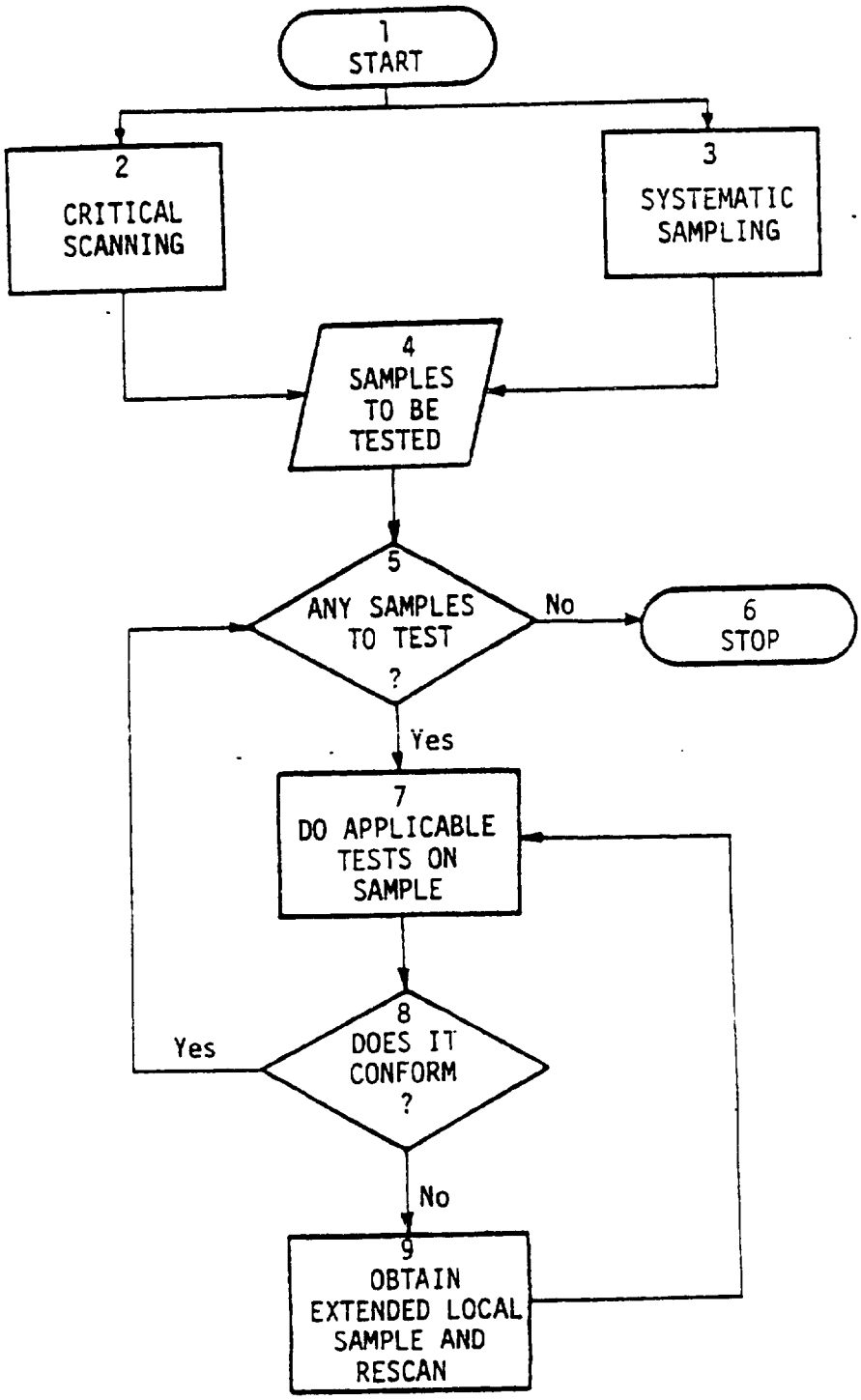


FIGURE 7. General logic of technical manual sampling plan.

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APPENDIX A

GUIDE TO PREFERRED VERB USAGE

10. GENERAL

10.1 Application. This appendix provides the data for implementation of 4.4.1.4.

10.2 Purpose. This appendix establishes a guide for standardizing verb usage and simplifying technical writing in technical manuals, especially in procedural statements.

20. REFERENCED DOCUMENTS

Not applicable.

30. DEFINITIONS

Not applicable.

40. GENERAL REQUIREMENTS

40.1 Preferred verb usage. Table I contains the data to be used in implementing 4.4.1.4.

50. DETAILED REQUIREMENTS

Not applicable.

TABLE I. Guide to preferred verb usage.

Verbs	Preferred alternatives	Examples and notes
Accomplish	Perform, do	
Actuate	Operate, move	
Adapt		(Use more specific verb)
Add		
Adjust		(Use more specific verb)
Advise	Report to, tell	
Advance	Move forward, Move ahead	
Agitate	Shake	
Aid	Help	
Alert	Warn	
Aline		Spelling: Preferred over align.
Allocate	Assign, distribute	
Allow	Let	Let the sediment settle.
	Wait for	Wait for the sediment to settle.
	Leave	Leave two inches of slack in the line.
Alternate	Go back and forth	
Analyze	Think about	(Use more specific verb)
Apply		(Use more specific verb)
Arrange	Put in order	
Ascertain	Be sure	
Assemble		
Assess	Evaluate	
Assign		
Assist	Help	
Assure	Tell, report to	
Attach		
Avoid		
Back off		Back off nut three turns.
Balance		
Be sure		
Be careful		
Bend		

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TABLE I. Guide to preferred verb usage. - Continued

Verbs	Preferred alternatives	Examples and notes
Bleed		Specialized usage: Take out or let out some or all of a contained substance. Bleed off tank air pressure.
Blow		
Break		
Buck		Specialized usage: Reseat or tighten rivets from the shank side.
Calculate		
Calibrate		
Cap		Cap the pipe.
Catch		
Categorize	Classify	
Center		
Change	Replace, modify	
Channel	Form, cut, or wear a groove.	
Charge		Charge the fire extinguisher. Charge the battery.
Check	Be sure	
Checkout		Definition: Perform specified operations to verify operational readiness of a subcomponent, component, subsystem, or system.
Choose		
Clamp		
Classify		
Clean		
Clear		Clear the passageway.
Close		
Coat		
Code		Color code the parts.
Collect		
Communicate	Report to, tell	
Compare		
Compile	Collect	
Comply	Follow	Follow directions exactly.
Compress		Compress the spring.
Compute	Calculate	
Condition		(Use more specific verb)
Confer	Ask	
Connect		
Consolidate		(Use more specific verb)
Construct	Make, build, assemble	(Use more specific verb)
Control		(Use more specific verb)
Coordinate		(Use more specific verb)
Copy		
Correct		
Cover		
Cut		
Deflate		
Depress	Press, push	
Depressurize	Release pressure	
Destroy		
Detect		(Use more specific verb)
Determine	Measure	Measure the fluid level.
	Be sure	Be sure that the hatch is closed.
Develop		(Use more specific verb)
Devise		(Use more specific verb)
Diagnose		(Use more specific verb)
Disassemble		
Dismantle	Disassemble	
Disconnect	Unplug	

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TABLE I. Guide to preferred verb usage. - Continued

Verbs	Preferred alternatives	Examples and notes
Disengage	Release, unlock	
Dispatch	Send	
Dispose of		
Distribute	Hand out, spread out	
Drain		
Dry		
Effect	Perform, do	
Eliminate	Get rid of	
Employ	Use	
Empty		
Enforce		(Use simpler and more specific verb)
Engage		
Ensure	Be sure	
Enter	Go in, come in write in, fill in	Write the measurements in blanks 1-5.
Erect	Set up, put together, assemble	(Use more specific verb)
Establish		
Estimate		
Evaluate		
Examine	Inspect	(Use more specific verb)
Expedite		
Extend	Stretch out, make longer	
Extract	Pull out	
Fabricate	Make, build	
Figure	Calculate	
File		
Fill		
Find	Measure	Measure the amount of tension on the cable.
Flush		
Fold		Follow directions.
Follow		(Use more specific verb)
Form		
Force		
Furnish	Give	
Go to		
Ground		
Guard		
Guide	Insert carefully	
Hand		(Use more specific verb)
Handle		
Help		(Use more specific verb)
Hold		
Identify		
Immerse	Submerge	
Improve		
Indicate	Point out, tell	
Inflate		
Inform	Tell, report to	
Initiate	Start, begin	
Inject	Force	
Insert		
Inspect		
Install		
Insure	Be sure	(Use more specific verb)
Intercept		
Interchange		(Use more specific verb)
Interpret	Explain	(Use more specific verb)
Investigate		

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TABLE I. Guide to preferred verb usage. - Continued

Verbs	Preferred alternatives	Examples and notes
Isolate		(Use more specific verb)
Join	Attach, connect	
Keep		
Kick		
Latch		
Leave		
Let		
Level		
Lift		
Light		
Listen		
Load		
Locate	Find	
Lock		
Look for		
Loop		
Loosen		
Lower		
Lubricate		
Maintain		(Use more specific verb)
Make		
Mark	Tag, if marking with a tag	
Mate	Attach, connect	
Measure		
Mix		
Modify		
Monitor	Watch	
Mount	Install, attach	
Move		
Neutralize		(Use more specific verb)
Notify	Report to, tell	
Observe	Watch	Watch the meter closely.
	Obey	Obey safety regulations.
Obtain	Get	
Open		
Operate		(Use more specific verb)
Order	Put in order, requisition	
Organize		(Use more specific verb)
Orient	Put, set	Set the instrument so both technicians can see it.
	Introduce	Introduce personnel to the new procedures.
Originate	Start	
Overhaul		
Pack		
Paint		
Patch		
Perform		
Place	Put, set	
Plan		
Plug		
Plug in		Refers to an electrical device.
Position	Put, set	
Prepare		(Use more specific verb)
Pre-set	Set	
Press		
Pressurize		
Prevent		
Probe		(Use simpler and more specific verb)
Process		(Use more specific verb)

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TABLE I. Guide to preferred verb usage. - Continued

Verbs	Preferred alternatives	Examples and notes
Provide	Give	
Pull		
Pump		
Puncture		(Use more specific verb)
Purge		
Push		
Put		
Raise		
Read		
Readjust	Adjust	(Use more specific verb)
Ready	Prepare	
Reassemble	Assemble	
Recap	Cap	
Receive	Get	(Use more specific verb)
Recondition		
Reconnect	Connect	
Record	Write, write in	
Reduce		(Use more specific verb)
Regulate		
Reinflate	Inflate	
Reinstall	Install	
Reject	Do not use, destroy, dispose of	
Relay	Give, tell (message)	
Release		Reduce pressure.
Relieve	Reduce	
Remove		(Use simpler and more specific verb)
Renovate		(Use more specific verb)
Repair		
Repeat		Put back tools after use.
Replace	Put back, Install new	Install new motor in fan.
Replenish		(Use simpler and more specific verb)
Report to	Tell	
Repressurize	Pressurize	
Request	Ask for	
Requisition		
Reset	Set	(Use simpler and more specific verb)
Resolve		
Retard	Hold back, slow down	
Retract	Pull back	
Return		(Use more specific verb)
Review		(Use more specific verb)
Rework		
Rinse		
Rope off		
Rotate	Turn	
Route	Send	
Rub		(Use more specific verb)
Safeguard		Safety lock nut on relief valve.
Safety		Safety wire the bolts.
Safety wire		(Use more specific verb)
Salvage		(Use more specific verb)
Scan	Watch	
Schedule		
Screw		Screw rod into mounting bracket, but fasten or install screws instead of screw screws.
Scrub		(Use more specific verb)
Secure		

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TABLE I. Guide to preferred verb usage. - Continued

Verbs	Preferred alternatives	Examples and notes
Select	Choose	
Service		(Use more specific verb)
Set		
Set up		(Use more specific verb)
Shake		
Shut down		Shut down the air conditioning.
Signal		
Simulate		(Use carefully-avoid confusion with stimulate)
Stimulate		(Use simpler verb)
Stop	Shut-down (with regard to equipment)	
Store		
Stow		
Strike	Hit	
Submit		
Superintend	Supervise	
Supervise		
Support	Hold up	
Survey		
Synchronize		
Tabulate	Make a table, list	
Tag		
Take		
Tap		
Tell		
Test		(Use more specific verb)
Throw (a switch)	Set switch to ...	(Use more specific verb)
Tie		
Tighten		
Tilt		
Torque		(Use torque instead of tighten when a measured amount of force is to be applied.)
Tow		
Trace		
Transfer	Move	
Transmit		
Transport	Move (by vehicle)	
Trim		(Use more specific verb)
Troubleshoot		(Use more specific verb)
Tune		(Use more specific verb)
Turn		
Turn off		
Turn on		
Uncap	Remove cap	
Unlock		
Unplug	Remove plug	
Unscrew	Remove screw	
Unwind		
Use		
Utilize	Use	
Verify	Be sure	
Wait		
Wash		
Watch		
Wire		(Use more specific verb)
Withdraw	Pull out	
Wrap		
Zero		Zero the ohm meter.

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APPENDIX B

READABILITY MEASURES

10. GENERAL

10.1 Application. The procedures described herein provide necessary detail for implementation of 5.3.1 and 5.3.2.

10.2 Purpose. The intent of these procedures is to furnish useful quantitative measures for two characteristics contributing to comprehensibility of text, namely, word length and sentence length. These procedures have been adapted for use with technical documentation.

20. REFERENCED DOCUMENTS

Not applicable.

30. DEFINITIONS

30.1 ASL. Average Sentence Length.

30.2 AWL. Average Word Length.

30.3 Ordinary word. A subset of words (see 30.5) which excludes:

(a) Any alphanumeric or symbolic combination.
(b) Symbols or variable abbreviations from technical fields such as mathematics, electronics, physics, chemistry, and engineering.
(c) Abbreviations and acronyms.
(d) Numbers larger than 100.
(e) Formulas.

30.4 Sentence. A group of words conveying a complete thought and terminated by a period, exclamation mark, or question mark. Paragraph and procedure headings are not considered sentences.

30.5 Word. Any string of characters (letters, numbers, or punctuation) separated from other strings by one or more spaces. Each part of a hyphenated word which would be a meaningful word if standing alone shall be counted as a separate word.

40. GENERAL REQUIREMENTS

Not applicable.

50. DETAILED REQUIREMENTS

50.1 Preparation of text. Mark words which are not ordinary words (see 30.3 and 30.5 of this appendix).

50.2 Counts.

50.2.1 Words. Count and record the number of words in each sentence, and add these counts to obtain the total word count. Count marked words (see 50.1) and subtract from the total word count to obtain the ordinary word count.

50.2.2 Sentences. Count and record the number of sentences.

50.2.3 Syllables. Count and record the total number of syllables, using only the ordinary (unmarked) words. Count syllables as the words are pronounced when carefully read aloud. Examples:

placed	= 1 syllable	(-ed is not a syllable)
dated	= 2 syllables	(-ed is a syllable)
factory	= 3 syllables	(not fact'ry, as usually slurred)

50.3 Automatic counting equipment. Devices for obtaining keystroke, word, and sentence counts automatically as text is typed may be used. Keystroke counts will then substitute for

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syllable counts, but the counts must be adjusted to eliminate paragraph and procedure headings and keystroke counts must be based only on ordinary words.

50.4 Calculation of ASL and AWL. ASL is calculated by dividing the total word count by the number of sentences. AWL is calculated by dividing the syllable or keystroke count by the ordinary word count.

50.5 Long sentences. Since the total word count was accumulated by sentence, excessively long sentences (see 5.3.3) can be counted from these data.

50.6 Counting technique. Obtaining accurate word and syllable counts rapidly can be aided by using a pushbutton operated counting device while reciting the text aloud. Certain hand-held electronic calculators can be used in this way.

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APPENDIX C

SAMPLING PLAN FOR INSPECTION OF TECHNICAL MANUALS

10. GENERAL

10.1 Scope. This sampling plan applies to all documents listed in 1.4 and shall be used for validation if so directed by the procuring activity. In addition, if directed by the procuring activity, it shall be used with available portions of the manuscript, as applicable, during technical manual production and for in-process reviews.

10.2 Need. A sampling plan is required because the number of characteristics to be checked makes 100 percent evaluation excessively time consuming and costly for many technical manuals. Simple random sampling of pages is not an adequate method because a technical manual is not a homogeneous mass, and because different tests require different types and amounts of sample material.

10.3 Rationale. The plan presented here capitalizes on a mixture of random sampling by type of material and human visual scanning capability to detect instances of nonconformance. It further makes the assumption that material surrounding an instance of nonconformance is the most likely place for other such instances to occur.

20. REFERENCED DOCUMENTS

Not applicable.

30. DEFINITIONS

30.1 Manuscript. A general term to refer to the technical material available for evaluation. As directed by the procuring activity, this may be any portion of the manual available for in-process review, or may be the completed manual to be delivered to the procuring activity.

40. GENERAL REQUIREMENTS

40.1 Derivation of samples. Figure 7 shows the general logic of the sampling process, but not necessarily the chronological sequence, since some parts of the process can be carried out concurrently. Samples to be tested (see figure 7, block 4) are obtained as a result of the following two processes:

40.1.1 Critical scanning. The purpose of critical scanning (see figure 7, block 2) is to locate instances of suspected nonconformance for specific measurement. Each page of the entire manuscript shall be scanned superficially for obvious nonconformance (see 50.1). Any material which is suspected of nonconformance shall be subjected to appropriate measurement procedures. At any time, if any material is supected of nonconformance with a requirement of this standard, appropriate measurements shall be applied to that material.

40.1.2 Systematic sampling. The purpose of systematic sampling (see figure 7, block 3) is to provide semi-random samples of different types of material throughout the manual for detailed examination and testing. Different tests require different types and amounts of sample material. In general, the method of deriving samples is to identify a number of equally spaced "break" pages and to define sample material with reference to these break pages (see 50.2).

40.2 Applicable tests. Sample material shall be examined or tested in detail for conformance with the requirements of this standard (see figure 7, block 7). In most instances, the nature of the tests is implicit in the statement of the requirements. This is not the case for certain readability requirements (see 5.3) and additional detail is provided in Appendix B. Because of the nature of the requirements, conformance with 4.1 and 4.2 cannot be evaluated on the basis of samples.

40.3 Extended local samples. Under the assumption that the material surrounding an instance of nonconformance has a higher probability of containing more such instances, the process of extended local sampling (see figure 7, block 9) shall provide additional sample material for examination. Any new instances of nonconformance in the extended local sample shall result in further extended sampling (see 50.3).

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50. DETAILED REQUIREMENTS

50.1 Scanning procedure. Each page of the manuscript shall be scanned superficially. Grounds for suspected nonconformance shall include, but shall not be limited to, the following criteria; alternative equivalent criteria shall be developed if the manuscript is not typewritten single column.

50.1.1 Scanning criteria for procedures. Scanning criteria for procedures are as follows:

(a) Set of procedural steps has no heading.
(b) Procedure has more than ten steps.
(c) Step is more than three typewritten lines.
(d) One sentence is longer than two lines.
(e) Procedural steps are prefaced by unnecessary lead-in which merely duplicates the heading.
(f) Procedure is not prefaced by relevant introductory information such as personnel required, special tools, and test equipment.

50.1.2 Scanning criteria for nonprocedural text. Scanning criteria for nonprocedural text are as follows:

(a) Paragraph exceeds 6 typewritten lines.
(b) Page has one paragraph heading or none.
(c) Paragraph heading appears vague.
(d) Paragraph heading is nearly a full line long.

50.1.3 Scanning criteria for illustrations. Scanning criteria for illustrations are as follows:

(a) Portions of image area appear cluttered:
    (1) Too many symbols.
    (2) Too many line intersections.
    (3) Too many lines on graph.
(b) Line width is uneven or inappropriate.
(c) Print seems too small.
(d) Callouts are hard to distinguish; arrangement or placement of callouts appears unacceptable.
(e) Signal flow (except for feedback) does not read from left to right or top to bottom.
(f) Locator view is required.
(g) Table appears crowded--no aids to staying in correct row or column.
(h) Use of halftone (photograph or shading).

50.2 Systematic sampling.

50.2.1 Identifying break pages. Break pages are pages which are equally spaced throughout the manuscript beginning with a randomly selected page. They are used as reference points to develop different kinds of samples as described in subsequent paragraphs.

(a) Randomly select a starting page, using a random number table, a computer random number generating function, or other acceptable random process.
(b) Find N, the number of break pages, as indicated below.

Number of pages in manuscript	N: Number of break pages
Under 26	3 minimum
26 to 90	5
91 to 150	8
151 to 280	13
281 to 500	20
501 to 1000	32
Over 1000	32 per 1000 pages

(c) Divide N into the number of pages in the manuscript to find the increment between break pages.
(d) Successively add the increment (keep two decimal places) to the starting page and substract the increment from the starting page to obtain the set of equally spaced break pages.

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50.2.2 Samples of procedural text. Beginning with each break page, the first complete procedure and subsequent complete procedures as needed shall be selected until the total number of procedural steps exceeds 20. For suspected nonconformance identified by critical scanning, the above sampling process shall be applied beginning with the suspected procedure.

50.2.3 Samples of nonprocedural text.

50.2.3.1 Calculation of AWL and ASL. Beginning with each break page, at least two paragraphs shall be selected including the first full paragraph of nonprocedural text and continuing to include full paragraphs until the total number of typed lines exceeds 20. For suspected nonconformance identified by critical scanning, material shall be selected as above, but beginning with the suspected paragraph.

50.2.3.2 Other tests on nonprocedural text. Six-page samples beginning with each break page shall be selected. For each instance of suspected nonconformance identified by critical scanning, a six-page sample surrounding each suspected instance shall be selected.

50.2.4 Samples of illustrations and tables. A sample of each type of illustration shall be obtained. The types are:

(a) Functional diagrams.
(b) Pictorials.
(c) Line graphs.

50.2.4.1 Sampling method for illustrations. Beginning with each break page, the first example encountered of each illustration type shall be selected (ignoring those already selected), no matter where in the manual it occurs. The intent of this procedure is to generate a sample for each graphic type no larger than the number of break pages, and to sample each graphic type adequately, whether the examples are spread out through the manual or are clustered together.

50.2.4.2 Tables. A sample of tables shall be selected by the same method.

50.3 Extended local samples.

50.3.1 Procedural text. Material in the vicinity of a confirmed instance of nonconformance shall be checked for similar nonconformance, using the two procedures immediately prior to the nonconforming procedure and the two procedures immediately following it. The provisions of this paragraph shall apply to any new instances of confirmed or suspected nonconformance.

50.3.2 Nonprocedural text.

50.3.2.1 Samples for ASL and AWL calculation. If nonconformance is found, ASL and AWL shall be calculated from the pages three pages prior to and three pages following the nonconforming material using the method of Appendix B. The process of adding and testing such adjacent samples shall continue as long as the adjacent samples fail to conform to the requirements.

50.3.2.2 Other tests on nonprocedural text. If nonconformance is found, material in the two six-page samples immediately adjacent to the nonconforming sample shall also be tested for conformance. The process of adding and testing adjacent samples shall continue as long as adjacent samples fail to conform to the requirements.

50.3.3 Illustrations and tables. Material in the vicinity of a confirmed instance of nonconformance shall be checked for similar nonconformance, using the two examples of the same illustration type or table immediately prior to the nonconforming example, and the two examples immediately following it. All illustrations and tables on the pages bound by these examples shall be rescanned for potential nonconformance of all kinds. The provisions of this paragraph shall apply to any new instances of confirmed or suspected nonconformance.

50.4 Special cases of sampling.

50.4.1 Short manuscripts. If there are less than six pages between break pages, use as samples all material between break pages in lieu of six-page samples previously specified.

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50.4.2 Material already sampled. For any particular test, if the material has already been sampled, the test results already obtained shall be used.

50.4.3 Shortage of material. If the amount of material called for does not exist, all available material shall be used.

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